

Lynbrook Math Summer Camp 2026

G4 Area Techniques in Geometry

Lynbrook Math Club

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G4.1 Lecture

Area is one of the most tested concepts in middle and high school competition math. While knowing your basic formulas (like $A = \frac{1}{2}bh$ for a triangle or $A = \pi r^2$ for a circle) is essential, AMC problems often require us to go a step further. We frequently have to find the areas of irregular shapes, or determine the ratio of areas between different figures without knowing their exact dimensions.

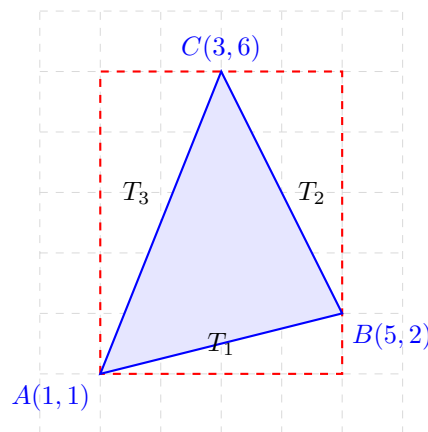
Finding Areas of Weird Shapes (The Subtractive Method)

Often, you will be asked to find the area of a polygon that doesn't fit a standard formula, or a shape on a coordinate plane. The best technique is the **Bounding Box** or **Subtractive Method**. Instead of finding the area directly, you enclose the weird shape in a standard shape (like a rectangle) and subtract the easy-to-calculate pieces on the outside.

Example 1

Find the area of $\triangle ABC$ with coordinates $A(1, 1)$, $B(5, 2)$, and $C(3, 6)$.

Solution: While you could use the Shoelace Theorem, the subtractive method is often faster and less prone to arithmetic errors. We draw a bounding rectangle around the triangle. The minimum x is 1, maximum x is 5. The minimum y is 1, maximum y is 6.



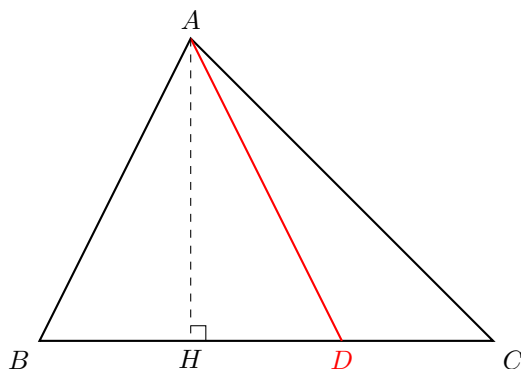
The bounding rectangle has width $5 - 1 = 4$ and height $6 - 1 = 5$, so its area is 20. There are three right triangles outside $\triangle ABC$ but inside the rectangle:

- Bottom triangle (T_1): base 4, height 1. Area = $\frac{1}{2}(4)(1) = 2$.
- Right triangle (T_2): base 2, height 4. Area = $\frac{1}{2}(2)(4) = 4$.
- Left triangle (T_3): base 2, height 5. Area = $\frac{1}{2}(2)(5) = 5$.

The area of $\triangle ABC$ is $20 - (2 + 4 + 5) = 9$.

Same Height Implies Ratio of Bases = Ratio of Areas

If two triangles share the same height (altitude), then the ratio of their areas is exactly the ratio of their bases. This most commonly happens when a line segment (a *cevian*) connects a vertex of a triangle to a point on the opposite side.



In the figure above, $\triangle ABD$ and $\triangle ADC$ share the exact same altitude AH . Therefore:

$$\frac{\text{Area}(\triangle ABD)}{\text{Area}(\triangle ADC)} = \frac{\frac{1}{2} \cdot BD \cdot AH}{\frac{1}{2} \cdot DC \cdot AH} = \frac{BD}{DC}$$

Example 2

In $\triangle ABC$, point D lies on BC such that $BD : DC = 2 : 3$. If the area of $\triangle ABC$ is 50, what is the area of $\triangle ABD$?

Solution: Because D splits the base BC in a $2 : 3$ ratio, the areas of $\triangle ABD$ and $\triangle ADC$ are also in a $2 : 3$ ratio. Let the area of $\triangle ABD$ be $2x$ and $\triangle ADC$ be $3x$. The total area is $2x + 3x = 5x = 50$, which means $x = 10$. Thus, the area of $\triangle ABD$ is $2(10) = 20$.

Similar Figures: Area Ratio is the Scale Factor Squared

If two shapes are similar with a scale factor of k (meaning every length in the second shape is k times the corresponding length in the first shape), then the ratio of their areas is k^2 .

Why? Because area is two-dimensional. If you multiply the base by k and the height by k , the area is multiplied by $k \times k = k^2$.

Example 3

Two regular hexagons have perimeters of 12 and 30. If the area of the smaller hexagon is 16, what is the area of the larger hexagon?

Solution: All regular hexagons are similar. The ratio of their perimeters is the same as the ratio of their side lengths, which is $30/12 = 5/2$. Because the linear scale factor is $k = 5/2$, the area scale factor is $k^2 = (5/2)^2 = 25/4$. The area of the larger hexagon is $16 \times \frac{25}{4} = \mathbf{100}$.